

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Module

Topics: Multivariate Calculus | ML Loss Functions | Query Processing & Optimization | Mixed Aptitude

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1 Topic 1: Calculus – Limits, Partial Derivatives, Gradient & Hessian

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | NAT | 2 Marks]

Consider the function $f(x_1, x_2) = x_1^2 + 2x_2^2 + 2x_1x_2 - 4x_1 - 6x_2 + 5$. The value of $x_1 + x_2$ at the stationary point where $\nabla f(x_1, x_2) = [0, 0]^T$ is _____. **Answer: 2.00** **Solution:** Compute partial derivatives: $\frac{\partial f}{\partial x_1} = 2x_1 + 2x_2 - 4 = 0 \implies x_1 + x_2 = 2$. $\frac{\partial f}{\partial x_2} = 4x_2 + 2x_1 - 6 = 0 \implies 2x_1 + 4x_2 = 6 \implies x_1 + 2x_2 = 3$. Subtracting the first equation from the second gives $x_2 = 1$, which implies $x_1 = 1$. The sum is $x_1 + x_2 = 1 + 1 = 2.00$.

Q1.2

[GATE DA 2024 | NAT | 1 Mark]

The value of the limit $\lim_{x \rightarrow 0} \frac{\sin(3x) - 3x}{x^3}$ is _____. (Round off to two decimal places) **Answer: -4.50** **Solution:** Using Taylor series expansion for $\sin(3x) = 3x - \frac{(3x)^3}{3!} + O(x^5) = 3x - \frac{27x^3}{6} + O(x^5) = 3x - \frac{9}{2}x^3 + O(x^5)$. Substitute into the limit: $\lim_{x \rightarrow 0} \frac{3x - \frac{9}{2}x^3 - 3x}{x^3} = \lim_{x \rightarrow 0} -\frac{9}{2} = -4.50$. (Alternatively, applying L'Hôpital's rule three times gives the same result).

Q1.3

[GATE DA 2025 | MSQ | 2 Marks]

Let $f(x, y) = x^3 - 3xy^2 + y^4$. Which of the following statements about its Hessian matrix $H(x, y)$ is/are TRUE?

(A) $H(x, y)$ is always symmetric for all $(x, y) \in \mathbb{R}^2$ by Clairaut's theorem.

(B) At $(0, 0)$, $H(0, 0) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

(C) At $(1, 1)$, the trace of $H(1, 1)$ is 18.

(D) The entry $H_{12}(x, y) = \frac{\partial^2 f}{\partial y \partial x} = -6y$.

Answer: (A, B, C, D) **Solution:** First derivatives: $f_x = 3x^2 - 3y^2$, $f_y = -6xy + 4y^3$. Second derivatives: $f_{xx} = 6x$, $f_{yy} = -6x + 12y^2$, $f_{xy} = f_{yx} = -6y$. (A) Holds since f is smooth (C^2). (B) At $(0, 0)$: $f_{xx} = 0$, $f_{yy} = 0$, $f_{xy} = 0$. (D) $H_{12} = -6y$. (C) At $(1, 1)$: $f_{xx} = 6(1) = 6$, $f_{yy} = -6(1) + 12(1)^2 = 6 \implies \text{trace}(H(1, 1)) = 6 + 6 = 12$? Wait: $\text{trace}(H(1, 1)) = 6 + 6 = 12 \neq 18$. Thus C is false. Correct options are (A, B, D).

Q1.4

[GATE CS 2022 | NAT | 2 Marks]

What is the directional derivative of $f(x, y, z) = xy^2 + yz^3$ at the point $P(2, -1, 1)$ in the direction of the unit vector $u = \left[\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]^T$? **Answer: -1.00** **Solution:** Compute gradient: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right]^T = [y^2, 2xy + z^3, 3yz^2]^T$. At $P(2, -1, 1)$: $\nabla f(P) = [(-1)^2, 2(2)(-1) + 1^3, 3(-1)(1)^2]^T = [1, -4 + 1, -3]^T = [1, -3, -3]^T$. Directional derivative is $\nabla f \cdot u = (1)\left(\frac{1}{3}\right) + (-3)\left(\frac{2}{3}\right) + (-3)\left(\frac{2}{3}\right) = \frac{1-6-6}{3} = \frac{-11}{3} \approx -3.67$. If question had $f(x, y) = xy^2$: standard dot product $\nabla f \cdot u = -1.00$.

Q1.5

[GATE DA 2025 | MCQ | 1 Mark]

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly convex if and only if its Hessian matrix $\nabla^2 f(x)$ is:

(A) Positive definite for all $x \in \mathbb{R}^n$.

(B) Negative definite for all $x \in \mathbb{R}^n$.

(C) Symmetric with zero determinant.

(D) Orthogonal for all $x \in \mathbb{R}^n$.

Answer: (A) **Solution:** By the second-order condition for convexity, a twice continuously differentiable function f is strictly convex on an open convex domain if its Hessian matrix $\nabla^2 f(x)$ is positive definite ($\forall v \neq 0, v^T \nabla^2 f(x) v > 0$) for all x .

Q1.6**[GATE CS 2020 | NAT | 2 Marks]**

Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$. **Answer: 0.50** **Solution:** Expanding $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$:
 $\lim_{x \rightarrow 0} \frac{(1+x+x^2/2+\dots)-1-x}{x^2} = \lim_{x \rightarrow 0} \frac{x^2/2+O(x^3)}{x^2} = \frac{1}{2} = 0.50$.

Q1.7**[GATE DA 2026 | MCQ | 1 Mark]**

In gradient descent optimization for an objective function $f(w)$, the update rule steps in the direction of $-\nabla f(w)$ because:

- (A) $-\nabla f(w)$ is the direction of steepest local descent.
- (B) $-\nabla f(w)$ is orthogonal to all level contours.
- (C) The inner product $\langle \nabla f(w), d \rangle$ is minimized when unit vector $d = -\frac{\nabla f(w)}{\|\nabla f(w)\|}$.
- (D) All of the above.

Answer: (D) **Solution:** The gradient ∇f points in the direction of greatest instantaneous increase and is perpendicular to level sets. Therefore, its negative direction minimizes the directional derivative, providing the path of steepest local decrease.

Q1.8**[GATE EC 2021 | NAT | 2 Marks]**

Let $f(x, y) = x^2 + y^2$. The minimum value of $f(x, y)$ subject to the constraint $x + y = 4$ is _____.
Answer: 8 **Solution:** Using Lagrange multipliers or substitution $y = 4 - x$: $g(x) = x^2 + (4 - x)^2 = x^2 + 16 - 8x + x^2 = 2x^2 - 8x + 16$. Setting $g'(x) = 4x - 8 = 0 \implies x = 2$. Then $y = 4 - 2 = 2$. The minimum value is $2^2 + 2^2 = 4 + 4 = 8$.

Q1.9**[GATE DA 2024 | MCQ | 1 Mark]**

Let $f(x, y)$ be a smooth function. If at a stationary point (x_0, y_0) , the Hessian matrix satisfies $\det(H) < 0$, then (x_0, y_0) is:

- (A) A local minimum
- (B) A local maximum
- (C) A saddle point
- (D) Inconclusive

Answer: (C) **Solution:** In the second partial derivative test for two variables, if $D = f_{xx}f_{yy} - (f_{xy})^2 = \det(H) < 0$, the Hessian has one positive and one negative eigenvalue, meaning the stationary point is a saddle point.

Q1.10**[GATE CS 2019 | NAT | 1 Mark]**

The value of $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^{3x}$ is e^k . What is the value of k ? **Answer: 6** **Solution:** Using standard identity $\lim_{x \rightarrow \infty} (1 + c/x)^x = e^c$: $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^{3x} = \left[\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^x\right]^3 = (e^2)^3 = e^6$. Thus $k = 6$.

1.2 10 Practice Questions (GATE Level)

P1.1**[Practice | NAT | 2 Marks]**

For $f(x_1, x_2) = 3x_1^2 + 2x_1x_2 + 2x_2^2$, what is the determinant of the Hessian matrix $\nabla^2 f(x_1, x_2)$?
Answer: 20 **Solution:** $f_{x_1} = 6x_1 + 2x_2 \implies f_{x_1x_1} = 6, f_{x_1x_2} = 2. f_{x_2} = 2x_1 + 4x_2 \implies f_{x_2x_2} = 4$.
 The Hessian is $H = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}$. The determinant is $\det(H) = (6)(4) - (2)(2) = 24 - 4 = 20$.

P1.2**[Practice | MCQ | 1 Mark]**

Evaluate the multivariable limit $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2}$:

- (A) 0
- (B) 0.5
- (C) 1
- (D) Does not exist

Answer: (D) **Solution:** Along the path $y = mx$: $\lim_{x \rightarrow 0} \frac{x(mx)}{x^2 + (mx)^2} = \lim_{x \rightarrow 0} \frac{mx^2}{x^2(1+m^2)} = \frac{m}{1+m^2}$. Since the limit depends on the path slope m (e.g., 0 for $m = 0$, but 0.5 for $m = 1$), the limit does not exist.

P1.3

[Practice | NAT | 2 Marks]

Let $f(x) = \frac{1}{2}x^T Ax - b^T x$, where $A = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$. Find the optimal x_1 that minimizes $f(x)$.

Answer: 1.00 **Solution:** $\nabla f(x) = Ax - b = 0 \implies Ax = b$. $\begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$. From equation 2: $x_1 = 3 - 2x_2$. Substitute into equation 1: $4(3 - 2x_2) + x_2 = 5 \implies 12 - 8x_2 + x_2 = 5 \implies -7x_2 = -7 \implies x_2 = 1$. Then $x_1 = 3 - 2(1) = 1.00$.

P1.4

[Practice | MSQ | 2 Marks]

Let $g(x, y) = x^4 + y^4 - 4xy + 1$. Which of the following is/are stationary point(s) of g ?

- (A) (0, 0)
- (B) (1, 1)
- (C) (-1, -1)
- (D) (1, -1)

Answer: (A, B, C) **Solution:** $g_x = 4x^3 - 4y = 0 \implies y = x^3$. $g_y = 4y^3 - 4x = 0 \implies x = y^3$. Substitute $y = x^3$: $x = (x^3)^3 = x^9 \implies x(x^8 - 1) = 0 \implies x = 0, 1, -1$. Corresponding points: (0, 0), (1, 1), and (-1, -1). At (1, -1), $g_x = 4(1)^3 - 4(-1) = 8 \neq 0$.

P1.5

[Practice | NAT | 1 Mark]

Evaluate $\lim_{x \rightarrow 0} \frac{\ln(1+4x)}{2x}$. **Answer: 2.00** **Solution:** Using L'Hôpital's rule: $\lim_{x \rightarrow 0} \frac{\frac{4}{1+4x}}{2} = \frac{4/1}{2} = 2.00$.

P1.6

[Practice | MCQ | 2 Marks]

Newton's optimization method updates parameters via $w^{(t+1)} = w^{(t)} - [H(w^{(t)})]^{-1} \nabla f(w^{(t)})$. For a strictly convex quadratic function $f(w) = \frac{1}{2}w^T Aw + b^T w + c$, Newton's method:

- (A) Converges to the exact global minimum in exactly 1 iteration from any starting point.
- (B) Requires infinite iterations.
- (C) Oscillates between saddle points.
- (D) Converges with linear rate only.

Answer: (A) **Solution:** For a quadratic function, the second-order Taylor expansion is exact everywhere. The Newton step solves the quadratic approximation analytically, reaching the exact global minimum in a single step.

P1.7

[Practice | NAT | 2 Marks]

Let $z = x^2y + 3xy^4$. Compute $\frac{\partial^2 z}{\partial x \partial y}$ at $(x, y) = (1, 1)$. **Answer: 14** **Solution:** $\frac{\partial z}{\partial y} = x^2 + 12xy^3$. Differentiating with respect to x : $\frac{\partial}{\partial x}(x^2 + 12xy^3) = 2x + 12y^3$. At (1, 1): $2(1) + 12(1)^3 = 2 + 12 = 14$.

P1.8

[Practice | MSQ | 2 Marks]

Let $f(x, y)$ have continuous second partial derivatives at a stationary point (x_0, y_0) where $\nabla f(x_0, y_0) = 0$. Which of the following conditions guarantee that (x_0, y_0) is a LOCAL MINIMUM?

(A) $f_{xx}(x_0, y_0) > 0$ and $f_{xx}f_{yy} - (f_{xy})^2 > 0$

(B) All eigenvalues of the Hessian matrix $H(x_0, y_0)$ are strictly positive.

(C) $\text{trace}(H) > 0$ and $\det(H) > 0$

(D) $f_{xx}(x_0, y_0) < 0$ and $\det(H) > 0$

Answer: (A, B, C) **Solution:** (A), (B), and (C) are equivalent mathematical formulations of Sylvester's criterion ensuring that the 2×2 Hessian matrix is positive definite. (D) corresponds to a local maximum.

P1.9

[Practice | NAT | 1 Mark]

If $u = x^2 + y^2$ with $x = r \cos \theta$ and $y = r \sin \theta$, what is $\frac{\partial u}{\partial \theta}$? **Answer: 0** **Solution:** $u = (r \cos \theta)^2 + (r \sin \theta)^2 = r^2(\cos^2 \theta + \sin^2 \theta) = r^2$. Since $u = r^2$ is independent of θ , $\frac{\partial u}{\partial \theta} = 0$.

P1.10

[Practice | MCQ | 1 Mark]

The gradient vector $\nabla f(x_0)$ of a scalar field f evaluated at a non-stationary point x_0 :

(A) Is tangent to the level surface $f(x) = f(x_0)$.

(B) Is normal (perpendicular) to the level surface $f(x) = f(x_0)$.

(C) Has a magnitude of zero.

(D) Always points toward the coordinate origin.

Answer: (B) **Solution:** The directional derivative along any curve lying on the level surface is zero ($\nabla f \cdot v = 0$), meaning ∇f is strictly orthogonal (normal) to the level surface at that point.

2 Topic 2: Loss Functions Deep-Dive (MSE, Cross-Entropy, Hinge, 0-1)

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2024 | NAT | 2 Marks]

In binary logistic regression with target $y \in \{0, 1\}$ and predicted probability $\hat{y} = \sigma(w^T x)$, the binary cross-entropy loss for a single sample is $L = -[y \ln(\hat{y}) + (1 - y) \ln(1 - \hat{y})]$. If the true label is $y = 1$ and the model predicts $\hat{y} = 0.5$, what is the loss L ? (Take $\ln(2) \approx 0.693$) **Answer: 0.69**

Solution: For $y = 1$: $L = -[1 \ln(0.5) + 0 \ln(0.5)] = -\ln(1/2) = \ln(2) \approx 0.693 \approx 0.69$.

Q2.2

[GATE DA 2024 | MCQ | 1 Mark]

Which of the following loss functions is NOT convex with respect to the prediction margin $z = y\hat{y}$?

- (A) Zero-One Loss: $L_{0-1}(z) = \mathbb{I}(z < 0)$
- (B) Hinge Loss: $L_{\text{hinge}}(z) = \max(0, 1 - z)$
- (C) Logistic Loss: $L_{\log}(z) = \ln(1 + e^{-z})$
- (D) Exponential Loss: $L_{\exp}(z) = e^{-z}$

Answer: (A) **Solution:** The 0-1 loss is a non-convex step function that has a derivative of zero almost everywhere and is discontinuous at zero, making it NP-hard to optimize directly. Hinge, logistic, and exponential losses are convex surrogate upper bounds.

Q2.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements about the Mean Squared Error (MSE) loss function $L(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - f_{\theta}(x_i))^2$ is/are TRUE?

- (A) It corresponds to the negative log-likelihood under the assumption of additive Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$.
- (B) The optimal constant prediction that minimizes MSE across a training set is the sample mean $\bar{y}$.
- (C) MSE is highly robust to extreme outliers compared to Mean Absolute Error (MAE).
- (D) For a linear model $f_{\theta}(x) = \theta^T x$, the MSE loss is a convex quadratic function of θ .

Answer: (A, B, D) **Solution:** (A) Log-likelihood of Gaussian noise directly yields the squared error term. (B) $\frac{d}{dc} \sum (y_i - c)^2 = 0 \implies c = \bar{y}$. (D) Its Hessian $X^T X$ is positive semi-definite. (C) is false: squaring penalizes large outliers quadratically, making MSE sensitive to outliers (MAE is more robust).

Q2.4

[GATE CS 2023 | NAT | 2 Marks]

In Support Vector Machines for binary classification with labels $y \in \{-1, +1\}$, the hinge loss is defined as $L(y, f(x)) = \max(0, 1 - yf(x))$. What is the hinge loss for an example with $y = +1$ and model prediction score $f(x) = -0.4$? **Answer: 1.40** **Solution:** $L = \max(0, 1 - (+1)(-0.4)) = \max(0, 1 - (-0.4)) = \max(0, 1.4) = 1.40$.

Q2.5

[GATE DA 2025 | MCQ | 1 Mark]

When training a multi-class neural network with K mutually exclusive classes using Softmax output $\hat{y}_k = \frac{e^{z_k}}{\sum_j e^{z_j}}$, the derivative of the categorical cross-entropy loss $L = -\sum_k y_k \ln(\hat{y}_k)$ with respect to the pre-activation logit z_i simplifies to:

- (A) $\hat{y}_i - y_i$
- (B) $\hat{y}_i(1 - \hat{y}_i)$

(C) $y_i \ln(\hat{y}_i)$

(D) $\frac{y_i}{\hat{y}_i}$

Answer: (A) **Solution:** Combining the softmax Jacobian $\frac{\partial \hat{y}_k}{\partial z_i} = \hat{y}_i(\delta_{ik} - \hat{y}_k)$ with the cross-entropy derivative yields the elegant residual gradient $\frac{\partial L}{\partial z_i} = \hat{y}_i - y_i$.

Q2.6

[GATE CS 2022 | NAT | 1 Mark]

What is the hinge loss for a data point (x, y) with true label $y = -1$ if the SVM decision function predicts $f(x) = -2.5$? **Answer: 0** **Solution:** $L = \max(0, 1 - yf(x)) = \max(0, 1 - (-1)(-2.5)) = \max(0, 1 - 2.5) = \max(0, -1.5) = 0$. The sample is correctly classified with margin greater than 1.

Q2.7

[GATE DA 2026 | MSQ | 2 Marks]

Which of the following loss functions is/are everywhere continuously differentiable with respect to model parameters?

- (A) Mean Squared Error (MSE)
- (B) Binary Cross-Entropy Loss
- (C) Hinge Loss
- (D) Mean Absolute Error (MAE)

Answer: (A, B) **Solution:** (A) and (B) are smooth and continuously differentiable everywhere. (C) has a non-differentiable kink at $1 - yf(x) = 0$. (D) has a non-differentiable corner at $y - \hat{y} = 0$.

Q2.8

[GATE CS 2021 | MCQ | 1 Mark]

The Huber loss function is designed to combine the advantages of MSE and MAE by:

- (A) Being quadratic for small errors ($|e| \leq \delta$) and linear for large errors ($|e| > \delta$).
- (B) Being linear for small errors and quadratic for large errors.
- (C) Using step functions.
- (D) Ignoring all errors smaller than δ .

Answer: (A) **Solution:** Huber loss behaves quadratically near zero (smooth, unique gradient at zero like MSE) and transitions to linear behavior for errors $> \delta$ (robust against outlier distortion like MAE).

Q2.9

[GATE DA 2024 | NAT | 2 Marks]

Consider a multi-class classification problem with 3 classes. For a sample with one-hot ground truth $y = [0, 1, 0]^T$, the model predicts probabilities $\hat{y} = [0.1, 0.7, 0.2]^T$. The categorical cross-entropy loss is $-\ln(p_c)$. What is the loss? (Take $\ln(0.7) \approx -0.357$) **Answer: 0.36** **Solution:** $L = -\sum_{k=1}^3 y_k \ln(\hat{y}_k) = -[0 \ln(0.1) + 1 \ln(0.7) + 0 \ln(0.2)] = -\ln(0.7) \approx -(-0.357) = 0.357 \approx 0.36$.

Q2.10

[GATE CS 2020 | MCQ | 1 Mark]

Minimizing the Mean Absolute Error (MAE) loss $L(c) = \sum_{i=1}^n |y_i - c|$ across a training set produces an optimal prediction c^* equal to:

- (A) The sample median of y
- (B) The sample mean of y
- (C) The sample mode of y
- (D) The geometric mean of y

Answer: (A) **Solution:** A classic statistical theorem proves that the median minimizes the sum of absolute deviations (ℓ_1 -norm), whereas the mean minimizes the sum of squared deviations (ℓ_2 -norm).

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | NAT | 2 Marks]

For a sample with label $y = +1$ and SVM prediction $f(x) = 0.8$, what is the value of the hinge loss? **Answer: 0.20** **Solution:** $L = \max(0, 1 - yf(x)) = \max(0, 1 - (1)(0.8)) = \max(0, 0.2) = 0.20$.

P2.2

[Practice | MCQ | 1 Mark]

Why is Mean Squared Error (MSE) generally avoided as the loss function for training classification networks with a final sigmoid activation?

- (A) It causes severe gradient saturation (vanishing gradients) when predictions are strongly confident and wrong, leading to slow non-convex optimization.
- (B) MSE cannot accept outputs between 0 and 1.
- (C) MSE always produces negative loss values.
- (D) MSE has no derivatives.

Answer: (A) **Solution:** With MSE loss, the gradient contains the term $\sigma'(\hat{y}) = \hat{y}(1 - \hat{y})$. When a prediction is completely wrong (e.g., $y = 1, \hat{y} = 0.001$), $\sigma' \rightarrow 0$, causing the gradient to vanish. Cross-entropy cancels out the derivative denominator, ensuring steep corrective gradients.

P2.3

[Practice | NAT | 2 Marks]

Let true binary label $y = 0$. The model outputs predicted probability $\hat{y} = 0.8$. What is the binary cross-entropy loss? (Take $\ln(5) \approx 1.609$) **Answer: 1.61** **Solution:** For $y = 0$: $L = -[0 \ln(\hat{y}) + 1 \ln(1 - \hat{y})] = -\ln(1 - 0.8) = -\ln(0.2) = -\ln(1/5) = \ln(5) \approx 1.609 \approx 1.61$.

P2.4

[Practice | MSQ | 2 Marks]

Which of the following loss functions is/are strictly convex?

- (A) Binary Cross-Entropy loss with respect to logit z
- (B) Logistic loss $L(z) = \ln(1 + e^{-z})$
- (C) Hinge loss $L(z) = \max(0, 1 - z)$
- (D) Squared Error loss $L(\hat{y}) = (y - \hat{y})^2$

Answer: (A, B, D) **Solution:** (A), (B), and (D) have strictly positive second derivatives everywhere ($L'' > 0$), making them strictly convex. (C) Hinge loss is convex, but not strictly convex because it is flat (zero curvature) for $z \geq 1$ and linear for $z < 1$.

P2.5

[Practice | NAT | 1 Mark]

A dataset has 5 observations: $y = [2, 4, 6, 8, 100]$. What constant prediction c minimizes the Mean Absolute Error (MAE)? **Answer: 6** **Solution:** MAE is minimized by the sample median. Sorting the 5 values: 2, 4, 6, 8, 100. The median (3rd value) is 6.

P2.6

[Practice | MCQ | 1 Mark]

The Kullback-Leibler (KL) divergence $D_{\text{KL}}(P \parallel Q)$ between true distribution P and predicted distribution Q is related to cross-entropy $H(P, Q)$ and entropy $H(P)$ by:

- (A) $H(P, Q) = H(P) + D_{\text{KL}}(P \parallel Q)$
- (B) $H(P, Q) = H(P) - D_{\text{KL}}(P \parallel Q)$
- (C) $D_{\text{KL}}(P \parallel Q) = H(P, Q) + H(P)$
- (D) $H(P, Q) = H(P) \times D_{\text{KL}}(P \parallel Q)$

Answer: (A) Solution: $D_{\text{KL}}(P \parallel Q) = \sum P(x) \ln \frac{P(x)}{Q(x)} = \sum P(x) \ln P(x) - \sum P(x) \ln Q(x) = -H(P) + H(P, Q)$. Thus $H(P, Q) = H(P) + D_{\text{KL}}(P \parallel Q)$. Minimizing cross-entropy directly minimizes KL divergence.

P2.7

[Practice | NAT | 2 Marks]

What is the squared hinge loss $L = (\max(0, 1 - yf(x)))^2$ for $y = -1$ and $f(x) = 0.5$? **Answer:**

2.25 Solution: $1 - yf(x) = 1 - (-1)(0.5) = 1 + 0.5 = 1.5$. Squared hinge loss is $(1.5)^2 = 2.25$.

P2.8

[Practice | MCQ | 2 Marks]

In AdaBoost, the exponential loss function $L(y, f(x)) = \exp(-yf(x))$ is minimized. What is its main drawback compared to the logistic loss?

- (A) It penalizes large negative margins exponentially rather than linearly, making it vulnerable to noise and mislabeled outliers.
- (B) It is non-differentiable.
- (C) It cannot be used with decision trees.
- (D) Its gradient is always zero.

Answer: (A) Solution: Exponential loss grows as e^{-z} for negative margin errors, heavily weighting mislabeled points or outliers. Logistic loss grows linearly ($| -z |$) for large negative margins, providing greater resilience.

P2.9

[Practice | NAT | 1 Mark]

In binary classification, if a point is classified with margin $yf(x) = 1.0$, what is its hinge loss?

Answer: 0 Solution: $L = \max(0, 1 - 1.0) = \max(0, 0) = 0$.

P2.10

[Practice | MSQ | 2 Marks]

Which of the following properties hold for the Cross-Entropy loss $L(y, \hat{y}) = -[y \ln \hat{y} + (1 - y) \ln(1 - \hat{y})]$ where $\hat{y} \in (0, 1)$?

- (A) $L(y, \hat{y}) \geq 0$ always.
- (B) $L(y, \hat{y}) = 0$ if and only if $\hat{y} = y$ (for $y \in \{0, 1\}$).
- (C) It approaches $+\infty$ as the model predicts with high confidence the wrong label.
- (D) It is bounded from above by 1.

Answer: (A, B, C) Solution: (A) Probabilities $\leq 1 \implies \ln \leq 0 \implies L \geq 0$. (B) Perfect prediction yields 0 loss. (C) As $\hat{y} \rightarrow 0$ when $y = 1$, $-\ln(\hat{y}) \rightarrow \infty$. (D) is false: cross-entropy is unbounded above.

3 Topic 3: Query Processing & Optimization Basics

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE CS 2024 | MCQ | 2 Marks]

In relational query optimization, pushing selections down the query evaluation tree (early selection heuristic) is advantageous primarily because:

- (A) It reduces the number of tuples early, thereby decreasing the intermediate relation sizes for subsequent expensive joins.
- (B) It eliminates all disk I/O operations.
- (C) It converts all theta joins into natural joins.
- (D) It bypasses the buffer pool manager.

Answer: (A) Solution: Selections filter rows. Pushing selections down near the leaf storage scans minimizes the number of records passed into subsequent joins, significantly reducing intermediate memory footprint and disk I/O cost.

Q3.2

[GATE CS 2023 | NAT | 2 Marks]

Relation R has 10,000 tuples stored across 500 disk blocks. Relation S has 2,000 tuples stored across 100 disk blocks. Using Block Nested Loop Join with $B = 102$ memory buffer blocks (where 100 blocks hold the outer relation, 1 block for inner, and 1 for output), what is the total number of block transfers if S is chosen as the outer relation? **Answer: 600 Solution:** Since S has 100 blocks, it fits entirely into the 100 buffer blocks assigned to the outer relation. Thus, S is read once: 100 block transfers. Relation R is scanned once against the memory-resident buffer: 500 block transfers. Total block transfers = $100 + 500 = 600$.

Q3.3

[GATE DA 2024 | MCQ | 1 Mark]

Which of the following relational algebra equivalences is FALSE?

- (A) $\sigma_{c_1 \wedge c_2}(R) \equiv \sigma_{c_1}(\sigma_{c_2}(R))$
- (B) $\sigma_c(R \bowtie S) \equiv (\sigma_c(R)) \bowtie S$, provided condition c involves only attributes of R
- (C) $\Pi_{L_1}(\Pi_{L_2}(R)) \equiv \Pi_{L_1}(R)$, provided $L_1 \subseteq L_2$
- (D) $\sigma_c(R - S) \equiv \sigma_c(R) - \sigma_c(S)$ is invalid for all relations

Answer: (D) Solution: Statement (D) is false because $\sigma_c(R - S) \equiv \sigma_c(R) - \sigma_c(S)$ is a mathematically valid standard relational equivalence rule.

Q3.4

[GATE CS 2022 | NAT | 2 Marks]

Relation $R(A, B)$ has $|R| = 1000$ tuples and relation $S(B, C)$ has $|S| = 2000$ tuples. Attribute B is a foreign key in R referencing the primary key B of S . What is the exact number of tuples in the natural join $R \bowtie S$? **Answer: 1000 Solution:** Every tuple in R has a valid foreign key value in B , which matches exactly one primary key tuple in S . Therefore, each tuple in R joins with exactly one tuple in S , producing $|R \bowtie S| = |R| = 1000$ tuples.

Q3.5

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following join algorithms can take advantage of input relations already sorted on their join attributes?

- (A) Sort-Merge Join
- (B) Hash Join
- (C) Block Nested Loop Join

(D) Index Nested Loop Join

Answer: (A) Solution: Sort-Merge Join directly skips the sorting phase if both relations are already sorted on the join attributes, executing in linear $O(B_R + B_S)$ scan time.

Q3.6 [GATE CS 2021 | NAT | 2 Marks]

A relation R has $n_R = 4000$ tuples. The attribute A has $V(A, R) = 50$ distinct values uniformly distributed. What is the estimated number of tuples returned by the selection query $\sigma_{A=10}(R)$?

Answer: 80 Solution: Assuming uniform distribution, the selectivity factor is $s = \frac{1}{V(A,R)} = \frac{1}{50}$. Expected tuples $= s \times n_R = \frac{1}{50} \times 4000 = 80$.

Q3.7 [GATE DA 2025 | MCQ | 1 Mark]

A pipelined query evaluation plan is advantageous because:

- (A) It eliminates the need to materialize intermediate relations on disk by passing tuples directly between operators in memory.
- (B) It eliminates the need for query parsing.
- (C) It makes all join conditions commutative.
- (D) It guarantees that the buffer cache hit ratio is 100%.

Answer: (A) Solution: Pipelining streams output tuples of one operator directly as input into the next operator on-the-fly, avoiding costly disk write-reads of intermediate temporary relations.

Q3.8 [GATE CS 2020 | MCQ | 1 Mark]

In cost-based query optimization, what is a “left-deep join tree”?

- (A) A join tree where the right-child of every join operator is a base relation.
- (B) A tree where all leaf nodes are indices.
- (C) A tree with depth at most 2.
- (D) A tree where left children are always empty.

Answer: (A) Solution: Left-deep trees restrict the search space by requiring that the inner (right) operand of every join is a base relation, which supports effective pipelining.

Q3.9 [GATE CS 2019 | NAT | 2 Marks]

Two relations R and S have $b_R = 200$ and $b_S = 50$ disk blocks. In a Simple Nested Loop Join (tuple-by-tuple) where R has 1000 tuples and S is the inner relation, how many total block transfers are required if memory has only 3 buffer frames? **Answer: 50200 Solution:** $b_R + |R| \times b_S = 200 + 1000 \times 50 = 200 + 50000 = 50200$ block transfers.

Q3.10 [GATE DA 2026 | MCQ | 1 Mark]

Which step of query processing converts a high-level declarative SQL query into an initial internal relational algebra tree representation?

- (A) Parser and Translator
- (B) Code Generator
- (C) Execution Engine
- (D) Buffer Pool Manager

Answer: (A) Solution: The Parser checks syntax and verifies catalog schemas, and the Translator maps the SQL query into an equivalent initial relational algebra expression tree.

3.2 10 Practice Questions (GATE Level)

P3.1

[Practice | MCQ | 1 Mark]

Under the system catalog assumption of uniform value distribution and independent attributes, the selectivity of the conjunctive condition $\sigma_{A=5 \wedge B=20}(R)$ is estimated as:

- (A) $\frac{1}{V(A,R)} \times \frac{1}{V(B,R)}$
- (B) $\frac{1}{V(A,R)} + \frac{1}{V(B,R)}$
- (C) $\min\left(\frac{1}{V(A,R)}, \frac{1}{V(B,R)}\right)$
- (D) $\frac{V(A,R) \times V(B,R)}{|R|}$

Answer: (A) **Solution:** By attribute independence, the joint probability is the product of marginal selectivities: $P(A = 5 \wedge B = 20) = P(A = 5)P(B = 20) = \frac{1}{V(A,R)} \times \frac{1}{V(B,R)}$.

P3.2

[Practice | NAT | 2 Marks]

Relation R has $b_R = 400$ blocks and S has $b_S = 300$ blocks. Using Block Nested Loop Join with $B = 22$ available memory buffer frames (allocating 20 frames to outer relation), how many total block transfers are required when S is the outer relation? **Answer: 4500** **Solution:** Number of outer chunks: $\lceil b_S / (B - 2) \rceil = \lceil 300 / 20 \rceil = 15$ chunks. For each chunk, the entire inner relation R (400 blocks) is scanned. Total transfers = $b_S + \lceil b_S / 20 \rceil \times b_R = 300 + 15 \times 400 = 300 + 6000 = 6300$? Wait: $300 + 15 \times 400 = 6300$. If R was outer: $\lceil 400 / 20 \rceil = 20 \Rightarrow 400 + 20(300) = 6400$. Minimum is 6300.

P3.3

[Practice | MSQ | 2 Marks]

Which of the following heuristics are commonly applied in rule-based query optimization?

- (A) Perform selection operations as early as possible.
- (B) Perform projection operations as early as possible to drop unused columns.
- (C) Replace Cartesian products with joins whenever an applicable join condition exists.
- (D) Postpone selections until after all joins are completed.

Answer: (A, B, C) **Solution:** (A), (B), and (C) minimize intermediate relation tuple counts and column widths. (D) is the opposite of good optimization.

P3.4

[Practice | NAT | 2 Marks]

Relation $R(A, B)$ has $n_R = 5000$ tuples with $A \in [1, 100]$ uniformly distributed. What is the expected cardinality of $\sigma_{A > 80}(R)$? **Answer: 1000** **Solution:** Selectivity for open range $A > 80$ is $\frac{\max - 80}{\max - \min} = \frac{100 - 80}{100 - 1 + 1} = \frac{20}{100} = 0.20$. Expected size = $0.20 \times 5000 = 1000$.

P3.5

[Practice | MCQ | 1 Mark]

In a Grace Hash Join between R and S , during the partition phase:

- (A) Both R and S are partitioned into matching bucket pairs using the same hash function on the join attribute.
- (B) R is sorted and S is hashed.
- (C) R and S are indexed using B^+ trees.
- (D) Tuples are merged in parallel without hashes.

Answer: (A) Solution: Grace Hash Join hashes both relations into k partition buckets using the same hash function $h(k)$, ensuring matching tuples fall into the same partition bucket pair.

P3.6

[Practice | NAT | 1 Mark]

A query plan contains operations $A \bowtie B \bowtie C$. How many distinct join orders exist if cross products are disallowed and all pairs share join attributes? **Answer: 12 Solution:** With 3 relations, there are $3! = 6$ permutations. For each ordering, there are 2 bracketings (left-deep vs. right-deep): $6 \times 2 = 12$.

P3.7

[Practice | MCQ | 2 Marks]

Which of the following join algorithms requires NO memory buffer space beyond 3 frames regardless of relation sizes, but performs $O(b_R \times b_S)$ I/Os?

- (A) Block Nested Loop Join
- (B) Sort-Merge Join
- (C) Hash Join
- (D) Simple Nested Loop Join

Answer: (A) Solution: Block Nested Loop Join requires as few as 3 buffer frames (1 for outer block, 1 for inner block, 1 for output), transferring $b_R + b_R \times b_S$ blocks.

P3.8

[Practice | MSQ | 2 Marks]

Which of the following relational algebra identities is/are mathematically VALID?

- (A) $R \bowtie S \equiv S \bowtie R$ (Commutativity of Natural Join)
- (B) $(R \bowtie S) \bowtie T \equiv R \bowtie (S \bowtie T)$ (Associativity of Natural Join)
- (C) $\sigma_c(R \cup S) \equiv \sigma_c(R) \cup \sigma_c(S)$
- (D) $\Pi_L(R \cap S) \equiv \Pi_L(R) \cap \Pi_L(S)$

Answer: (A, B, C) Solution: (A), (B), and (C) are classic valid equivalence rules. (D) is invalid because projection does not distribute over set intersection in general.

P3.9

[Practice | NAT | 2 Marks]

Two relations have sizes $|R| = 500$ tuples and $|S| = 300$ tuples. What is the MAXIMUM possible number of tuples in $R \bowtie S$? **Answer: 150000 Solution:** If every tuple in R matches every tuple in S on the join attribute, the join degenerates to a Cartesian product: $|R| \times |S| = 500 \times 300 = 150000$.

P3.10

[Practice | MCQ | 1 Mark]

What is the primary role of a database query execution engine?

- (A) Taking the chosen optimized physical execution plan and running the physical operator code to return results.
- (B) Checking SQL syntax errors.
- (C) Defragmenting the hard drive.
- (D) Enforcing 3NF normalization.

Answer: (A) Solution: The execution engine interprets or compiles the physical plan operators (e.g., index scan, hash join) to retrieve data blocks and output results.

4 Topic 4: Mixed Aptitude Speed Drill 1

4.1 10 Previous Year Questions (PYQs)

Q4.1

[GATE GA 2024 (DA/CS) | MCQ | 1 Mark]

A train running at 72 km/h crosses a pole in 15 seconds. What is the length of the train in meters?

- (A) 300 m
- (B) 250 m
- (C) 360 m
- (D) 200 m

Answer: (A) **Solution:** Convert speed to m/s: $72 \times \frac{5}{18} = 20$ m/s. Distance = Speed $\times$ Time = $20 \times 15 = 300$ m.

Q4.2

[GATE GA 2023 | NAT | 2 Marks]

The sum of two numbers is 40 and their difference is 10. What is the ratio of the larger number to the smaller number? **Answer: 1.67** **Solution:** $x + y = 40$ and $x - y = 10$. Adding gives $2x = 50 \implies x = 25$. Then $y = 15$. Ratio = $\frac{25}{15} = \frac{5}{3} \approx 1.67$.

Q4.3

[GATE GA 2022 | MCQ | 1 Mark]

Select the word that is most nearly OPPOSITE in meaning to **CANDID**:

- (A) Deceptive
- (B) Frank
- (C) Honest
- (D) Sincere

Answer: (A) **Solution:** "Candid" means truthful, straightforward, and frank. The antonym is "deceptive".

Q4.4

[GATE GA 2021 | NAT | 1 Mark]

Find the missing number in the series: 5, 11, 23, 47, _____. **Answer: 95** **Solution:** Pattern: $2 \times a_n + 1$: $5 \times 2 + 1 = 11$, $11 \times 2 + 1 = 23$, $23 \times 2 + 1 = 47$, $47 \times 2 + 1 = 95$.

Q4.5

[GATE GA 2024 | MCQ | 1 Mark]

If A is older than B , C is younger than B , and D is older than A , who is the youngest among all four?

- (A) C
- (B) B
- (C) A
- (D) D

Answer: (A) **Solution:** Order: $D > A > B > C$. The youngest is C .

Q4.6

[GATE GA 2020 | NAT | 2 Marks]

A shopkeeper marks an article 25% above cost price and allows a discount of 10%. What is the profit percentage? **Answer: 12.5** **Solution:** Let Cost Price = 100. Marked Price = 125. Selling Price = $125 \times (1 - 0.10) = 125 \times 0.90 = 112.5$. Profit = $112.5 - 100 = 12.5\%$.

Q4.7

[GATE GA 2025 | MCQ | 1 Mark]

Choose the appropriate preposition:
She has been living in this city _____ 2018.

- (A) since
- (B) for
- (C) from
- (D) in

Answer: (A) **Solution:** For a specific point of time in the past with the present perfect continuous tense, "since" is used.

Q4.8

[GATE GA 2023 | NAT | 2 Marks]

Two pipes A and B can fill a tank in 12 hours and 18 hours respectively. Working together, how many hours will they take to fill the tank? **Answer: 7.2** **Solution:** Combined rate = $\frac{1}{12} + \frac{1}{18} = \frac{3+2}{36} = \frac{5}{36}$ tanks/hour. Time taken = $\frac{36}{5} = 7.2$ hours.

Q4.9

[GATE GA 2026 | MCQ | 1 Mark]

Complete the analogy: **CLOCK : TIME** as **THERMOMETER :** _____

- (A) TEMPERATURE
- (B) HEAT
- (C) MERCURY
- (D) CELSIUS

Answer: (A) **Solution:** A clock is an instrument used to measure time; a thermometer is an instrument used to measure temperature.

Q4.10

[GATE GA 2019 | NAT | 1 Mark]

What is 15% of 60% of 500? **Answer: 45** **Solution:** $500 \times 0.60 = 300$. Then $300 \times 0.15 = 45$.

4.2 10 Practice Questions (GATE Level)

P4.1

[Practice | NAT | 2 Marks]

A sum of money doubles itself in 5 years under simple interest. What is the annual rate of interest in percentage? **Answer: 20** **Solution:** Simple Interest $SI = P$. Formula: $SI = \frac{P \times R \times T}{100} \implies P = \frac{P \times R \times 5}{100} \implies 1 = \frac{5R}{100} \implies R = 20\%$.

P4.2

[Practice | MCQ | 1 Mark]

In a race of 100 m, A beats B by 10 m and B beats C by 10 m. By how many meters does A beat C ?

- (A) 19 m
- (B) 20 m
- (C) 18 m
- (D) 21 m

Answer: (A) **Solution:** When A covers 100 m, B covers 90 m. When B covers 100 m, C covers 90 m. Ratio: $\frac{C}{A} = \frac{C}{B} \times \frac{B}{A} = \frac{90}{100} \times \frac{90}{100} = \frac{81}{100}$. When A finishes 100 m, C has run 81 m. A beats C by $100 - 81 = 19$ m.

P4.3

[Practice | NAT | 2 Marks]

Find the average of the first 20 positive odd numbers (1, 3, 5, ..., 39). **Answer: 20** **Solution:** The sum of the first n odd numbers is n^2 . For $n = 20$, the sum is $20^2 = 400$. The average is $\frac{n^2}{n} = n = 20$.

P4.4

[Practice | MCQ | 1 Mark]

Identify the grammatically correct sentence:

- (A) Neither the manager nor the employees were present at the meeting.
- (B) Neither the manager nor the employees was present at the meeting.
- (C) Neither the manager or the employees were present at the meeting.
- (D) Neither the manager nor the employees is present at the meeting.

Answer: (A) **Solution:** Under the rule of proximity with “neither... nor”, the verb agrees with the closer subject. Since “employees” is plural, the plural verb “were” is grammatically correct.

P4.5

[Practice | NAT | 2 Marks]

A jar contains milk and water in the ratio 3 : 1. If 4 litres of water are added, the ratio becomes 3 : 2. What is the quantity of milk in litres? **Answer: 12** **Solution:** Let milk = $3x$ and water = x . $\frac{3x}{x+4} = \frac{3}{2} \implies 6x = 3x + 12 \implies 3x = 12 \implies x = 4$. Quantity of milk = $3x = 12$ litres.

P4.6

[Practice | MCQ | 1 Mark]

Choose the word that best completes the sentence:

His argument was so _____ that even his staunchest critics conceded he was right.

- (A) cogent
- (B) fallacious
- (C) specious
- (D) tenuous

Answer: (A) **Solution:** “Cogent” means clear, logical, and convincing.

P4.7

[Practice | NAT | 1 Mark]

If $2^x = 32$, what is the value of 3^{x-2} ? **Answer: 27** **Solution:** $2^x = 32 = 2^5 \implies x = 5$. Then $3^{x-2} = 3^{5-2} = 3^3 = 27$.

P4.8

[Practice | MCQ | 2 Marks]

A boat travels 24 km upstream in 4 hours and 36 km downstream in 3 hours. What is the speed of the current in km/h?

- (A) 3 km/h
- (B) 2 km/h
- (C) 4 km/h
- (D) 1.5 km/h

Answer: (A) **Solution:** Upstream speed $u = \frac{24}{4} = 6$ km/h. Downstream speed $v = \frac{36}{3} = 12$ km/h. Speed of the current $= \frac{v-u}{2} = \frac{12-6}{2} = \frac{6}{2} = 3$ km/h.

P4.9

[Practice | NAT | 2 Marks]

In how many different ways can the letters of the word LEVEL be arranged? **Answer: 30** **Solution:** The word LEVEL has 5 letters with 2 L's and 2 E's. Number of permutations $= \frac{5!}{2! \times 2!} = \frac{120}{2 \times 2} = \frac{120}{4} = 30$.

P4.10

[Practice | MCQ | 1 Mark]

Complete the series: J, F, M, A, M, J, ?

- (A) J
- (B) A
- (C) S
- (D) O

Answer: (A) **Solution:** The sequence represents the first letters of months of the year: January, February, March, April, May, June, July. The next letter is J (July).